

CSE 5830: Probabilistic Graph Analysis

Homework 3

Due 11:59PM, Wednesday 9/30/2026

Show all your work to get a full score.

Problem 1: Noisy-AND (15 points)

A Noisy-AND model (independence of the causal mechanisms when all X s are 1) with binary random variables and leaky parameters $(\lambda_0, \lambda_1, \dots, \lambda_K)$.

(a) Expression for $P(Y = 0 \mid X_1, \dots, X_K)$ (10 points)

Setup / convention. Noisy-AND is the De Morgan dual of noisy-OR. In a deterministic AND gate, a single input $X_i = 0$ is enough to force $Y = 0$, and $Y = 1$ happens in only one way: every input equals 1.

X_1	X_2	$Y = X_1 \wedge X_2$
0	0	0
0	1	0
1	0	0
1	1	1

For each parent, introduce an independent mechanism variable Z_i , where $Z_i = 1$ means “ X_i forces $Y = 0$ ”. An input that is present can never force an AND gate to 0, so

$$P(Z_i = 1 \mid X_i = 1) = 0, \quad P(Z_i = 1 \mid X_i = 0) = \lambda_i.$$

That is, λ_i is the probability that an absent input ($X_i = 0$) succeeds in forcing $Y = 0$. The leak Z_0 is always active, with $P(Z_0 = 1) = \lambda_0$: the probability that $Y = 0$ even when every input is 1. Then $Y = 1$ if and only if $Z_0 = Z_1 = \dots = Z_K = 0$.

Derivation. Since $Y = 1$ requires that no mechanism fires, and the mechanisms are independent (independence of the causal mechanisms),

$$P(Y = 1 \mid X_1, \dots, X_K) = P(Z_0 = 0) \prod_{i=1}^K P(Z_i = 0 \mid X_i).$$

Each factor is

$$P(Z_i = 0 \mid X_i) = \begin{cases} 1, & X_i = 1, \\ 1 - \lambda_i, & X_i = 0, \end{cases}$$

so parents with $X_i = 1$ contribute a factor of 1 and drop out. Hence

$$P(Y = 1 \mid X_1, \dots, X_K) = (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i),$$

and taking the complement,

$$P(Y = 0 \mid X_1, \dots, X_K) = 1 - (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i)$$

Sanity check. With no leak ($\lambda_0 = 0$) and $\lambda_i = 1$ for all i : if every $X_i = 1$, the product is empty (equal to 1), so $P(Y = 0) = 0$; if any $X_i = 0$, the product contains a factor $1 - 1 = 0$, so $P(Y = 0) = 1$. This is exactly the deterministic AND gate.

(b) Numerical case (5 points)

$K = 4$, $\lambda_0 = 0$, $\lambda_1 = 0.9$, $\lambda_2 = 0.75$, $\lambda_3 = 0.8$, $\lambda_4 = 0.85$.

Only X_2 and X_3 are off, so only λ_2 and λ_3 enter the product; $X_1 = X_4 = 1$ contribute factors of 1. With no leak, $1 - \lambda_0 = 1$. Using the formula from (a):

$$\begin{aligned} P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 0, X_4 = 1) &= 1 - (1 - \lambda_0)(1 - \lambda_2)(1 - \lambda_3) \\ &= 1 - (1 - 0)(1 - 0.75)(1 - 0.8) \\ &= 1 - (1)(0.25)(0.2) \\ &= 1 - 0.05 \\ &= \boxed{0.95} \end{aligned}$$

Sanity check. Two inputs are off, and each has a high chance (0.75 and 0.8) of forcing $Y = 0$. So $Y = 1$ requires both of those mechanisms to fail, which happens with probability only $0.25 \times 0.2 = 0.05$. A value of $P(Y = 0)$ close to, but below, 1 is what we expect.

Problem 2: Factor product (15 points)

Given factors:

X	Y	$\phi_1(X, Y)$	Y	Z	$\phi_2(Y, Z)$
1	1	0.8	1	1	0.2
1	2	0.5	1	2	0.2
2	1	0.5	2	1	0.9
2	2	0.6	2	2	1.0

(a) $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$ (10 points)

X	Y	Z	$\phi_1(X, Y) \cdot \phi_2(Y, Z)$	$\psi(X, Y, Z)$
1	1	1	$0.8 \cdot 0.2$	0.16
1	1	2	$0.8 \cdot 0.2$	0.16
1	2	1	$0.5 \cdot 0.9$	0.45
1	2	2	$0.5 \cdot 1.0$	0.5
2	1	1	$0.5 \cdot 0.2$	0.1
2	1	2	$0.5 \cdot 0.2$	0.1
2	2	1	$0.6 \cdot 0.9$	0.54
2	2	2	$0.6 \cdot 1.0$	0.6

(b) $\psi(X, Z \mid Y = 1)$ (5 points)

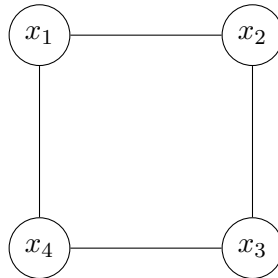
This is a factor reduction to the context $Y = 1$: keep only the rows of $\psi(X, Y, Z)$ from (a) with $Y = 1$ and drop the Y column. Reduction alone does not normalize, so the unnormalized values are shown first. To read $\psi(X, Z \mid Y = 1)$ as a conditional distribution, divide each entry by their sum, $0.16 + 0.16 + 0.1 + 0.1 = 0.52$.

X	Z	$\psi[Y = 1]$ (reduced)	$\psi(X, Z \mid Y = 1)$ (normalized)
1	1	0.16	$0.16/0.52 \approx 0.308$
1	2	0.16	$0.16/0.52 \approx 0.308$
2	1	0.1	$0.1/0.52 \approx 0.192$
2	2	0.1	$0.1/0.52 \approx 0.192$
Sum		0.52	1

Problem 3: Pairwise Markov network (10 points)

$$p(\mathbf{x}) \propto \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)$$

(a) Draw the network (2 points)



(b) Conditionals in terms of ϕ (8 points)

[TODO: Show the derivation once in full for $p(x_1 \mid x_2, x_4)$ (write the conditional as a ratio of joints, then cancel), then use the same pattern for the rest.]

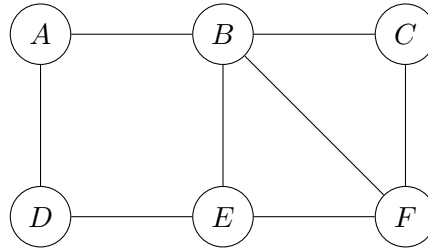
$$p(x_1 \mid x_2, x_4) = [\text{TODO: }]$$

$$p(x_2 \mid x_1, x_3) = [\text{TODO: }]$$

$$p(x_3 \mid x_2, x_4) = [\text{TODO: }]$$

$$p(x_4 \mid x_1, x_3) = [\text{TODO: }]$$

Problem 4: Markov network structure (20 points)



(a) All factors that factorize over the graph (6 points)

Edges: **[TODO: list them]**

Cliques: **[TODO: list every clique (single nodes, edges, triangles, ...)]**

Maximal cliques: **[TODO:]**

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of factors}]$$

(b) Gibbs distribution as a pairwise MN (6 points)

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of pairwise factors}], \quad Z = [\text{TODO: }]$$

(c) Local Markov independencies (6 points)

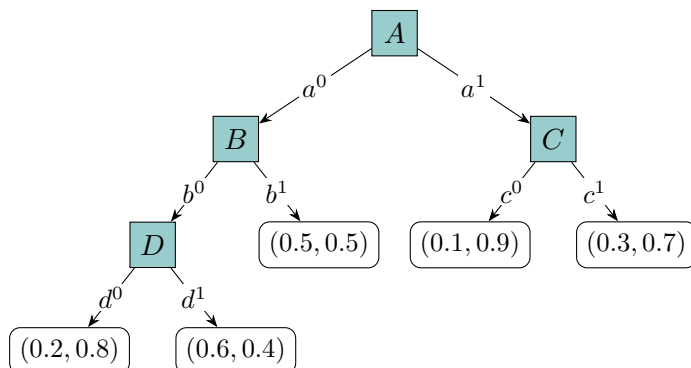
Node	Neighbors (Markov blanket)	Local independence
A	_____	[TODO:]
B	_____	[TODO:]
C	_____	[TODO:]
D	_____	[TODO:]
E	_____	[TODO:]
F	_____	[TODO:]

(d) Is $A \perp F \mid \{D, C\}$? (2 points)

[TODO: Answer yes/no, and justify with separation: either exhibit an active path from A to F that avoids {D, C}, or argue that every path passes through {D, C}.]

Problem 5: Tree-structured CPD (10 points)

Leaves list $(P(e^0), P(e^1))$.



(a) $P(E = e^0 \mid a^0, b^1, c^1, d^1)$

Path through the tree: **[TODO:]** Answer: _____

(b) $P(E = e^1 \mid a^1, b^1, c^1, d^1)$

Path through the tree: **[TODO:]** Answer: _____

(c) $P(E = e^1 \mid a^0, b^0, c^0, d^0)$

Path through the tree: **[TODO:]** Answer: _____

(d) Is $E \perp_c B \mid a^0$?

[TODO: Yes/no + justification using the tree.]

(e) Is $E \perp_c C \mid d^1$?

[TODO: Yes/no + justification using the tree.]

Use of Ai

In this assignment, I used Ai to help me understand the assignment give me hints for each question as well as helping whenever I got stuck in a question. Also used AI to write and debug my tex code. However, I solved all the question of the assignment on my own.